Design, Fabrication and Analysis of a Mach-Zehnder Interferometer

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Abstract: In this online course project, Mach-Zehnder Interferometers (MZI) are designed and fabricated using several different imbalance lengths. The difference in imbalance length and its effect on the free-spectral range (FSR) is investigated. Waveguide mode simulations using the meow FDE solver and system-level circuit simulations using SAX are used to predict the MZI transfer function. Corner analysis quantifies the effect of manufacturing variability (±20 nm in width, ±10 nm in height) on the transfer function. Numerical curve fitting extracts waveguide parameters (effective index, group index, dispersion) from the experimental data. Devices are fabricated on a SOI substrate designed to operate near 1550 nm.

Index Terms: Mach-Zehnder Interferometer (MZI), Silicon Photonics, Strip waveguides, Free-spectral range, Group index, Corner analysis.

1. Introduction

Silicon Photonic Circuits (PIC) have contributed to the rise of optical communications by combining the speed and compactness of photonics with standardised CMOS fabrication. The basic building blocks of a PIC are grating couplers, Y-branch splitters, and strip waveguides. These components are combined to form functional devices such as the Mach-Zehnder Interferometer (MZI), which is the subject of this report.

In this project, unbalanced MZIs with nine different path length differences (ΔL = 30 … 570 µm) are measured. The FSR of each device is related to the waveguide group index n_g and ΔL through

$$\mathrm{FSR} = \frac{\lambda^2}{n_g \cdot \Delta L}. \tag{1}$$

The objective is to extract the waveguide parameters by comparing simulation results with experimental measurements and to quantify the effect of manufacturing variability through corner analysis.

2. Theoretical Background

2.1 Y-branch splitter and combiner

The Y-branch splits the input intensity equally:

$$I_1 = I_2 = \frac{I_{\mathrm{in}}}{2}, \quad E_1 = E_2 = \frac{E_{\mathrm{in}}}{\sqrt{2}}. \tag{2}$$

When combining, the output intensity is:

$$I_{\mathrm{out}} = \frac{1}{2}(I_1 + I_2), \quad E_{\mathrm{out}} = \frac{1}{\sqrt{2}}(E_1 + E_2). \tag{3}$$

2.2 Waveguides

The complex propagation constant of a waveguide mode is:

$$\beta = 2\pi \cdot n_{\mathrm{eff}} \frac{\lambda}{\lambda} - i\frac{\alpha}{2}, \tag{4}$$

where n_{eff} is the effective index and α is the field attenuation coefficient. The effective index is expanded in a Taylor series about the centre wavelength λ_0:

$$n_{\mathrm{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \tag{5}$$

The group index is:

$$n_g(\lambda) = n_{\mathrm{eff}} - \lambda \frac{dn_{\mathrm{eff}}}{d\lambda} = n_1 - \lambda_0 n_2. \tag{6}$$

2.3 MZI transfer function

The power transmission of the unbalanced MZI in dB is:

$$T(\lambda) = 10 \log_{10}\left[\frac{1}{2}\left(1 + e^{-\alpha \Delta L} \cos\varphi\right)\right] + b, \tag{7}$$

where $\varphi = \left(2\frac{\pi}{\lambda}\right) n_{\mathrm{eff}} \Delta L$ and b is a vertical offset accounting for excess loss.

2.4 Free spectral range

Substituting Equation 6 into Equation 1 gives the FSR at wavelength λ:

$$\mathrm{FSR}(\lambda) = \frac{\lambda^2}{n_g(\lambda) \cdot \Delta L}. \tag{8}$$

For a waveguide with low dispersion ($n_3 \approx 0$) the FSR is nearly constant across the measurement bandwidth.

3. Design

3.1 Waveguide geometry

The waveguide is a 500 nm wide × 220 nm thick silicon strip on a SiO_2 buried oxide layer with SiO_2 top cladding, operating in the quasi-TE fundamental mode at a centre wavelength of 1550 nm.

3.2 MZI devices

Nine MZI devices were designed with imbalance lengths from 30 µm to 570 µm. The expected FSR at 1550 nm using the simulated group index $n_g \approx 4.21$ and the measured n_g at $\lambda = 1550$ nm (curve fit) are shown in Table 1.

ΔL (µm)	Expected FSR (nm)	Measured n_g
30	19.0	4.186
40	14.3	4.179
60	9.5	4.188
80	7.1	4.187
115	5.0	4.189
200	2.9	4.189
300	1.9	4.190
400	1.4	4.191
570	1.0	4.189

Table 1: MZI device parameters at $\lambda = 1550$ nm: expected FSR from meow n_g and measured n_g (curve fit).

3.3 Chip layout

The chip layout for the fabricated devices is shown in Figure 1. The chip contains grating couplers for vertical fibre coupling and straight calibration waveguides adjacent to each MZI device.

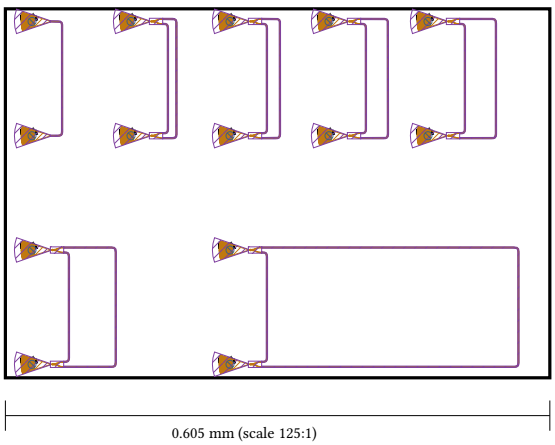

Figure 1: Chip layout (EBeam_anatolyb.gds, scale 125:1). Devices shown: calibratioion wavguide, MZI with ΔL = 30, 40, 80, 115, and 570 µm.

4. Simulation

4.1 Waveguide mode simulation

The fundamental quasi-TE mode of the 500 × 220 nm strip waveguide is simulated using the meow finite-difference eigenmode (FDE) solver [1] with Palik/Malitson dispersion data for Si and SiO_2 [2]. The simulation region is 2.5 × 1.7 µm with a non-uniform 100 × 100 mesh (60 % of points concentrated within 0.5 µm of the core) and 8 PML layers.

The effective index n_{eff} is obtained directly from the FDE eigenvalue. The group index is extracted using the electromagnetic energy-velocity relation [3]:

$$n_g = \frac{c(W_e + W_m)}{2P_z}, \tag{9}$$

where $P_z = \int \mathrm{Re}(\boldsymbol{E} \times \boldsymbol{H}^*) \cdot \hat{z}, dA$ is the Poynting power flux and the stored energy densities are

$$W_e = \varepsilon_0 \sum_i \int \left(n_i^2 - 2n_i \lambda \frac{\partial n_i}{\partial \lambda} \right) |E_i|^2 \, dA, \tag{10}$$

$$W_m = \mu_0 \sum_i \int |H_i|^2 \, dA. \tag{11}$$

The material dispersion term $\frac{\partial n}{\partial \lambda}$ in W_e is evaluated by finite difference ($\Delta\lambda = 1$ nm), making Equation 9 valid for dispersive materials without requiring a full frequency sweep. The results over the measurement bandwidth 1460–1580 nm are shown in Figure 2.

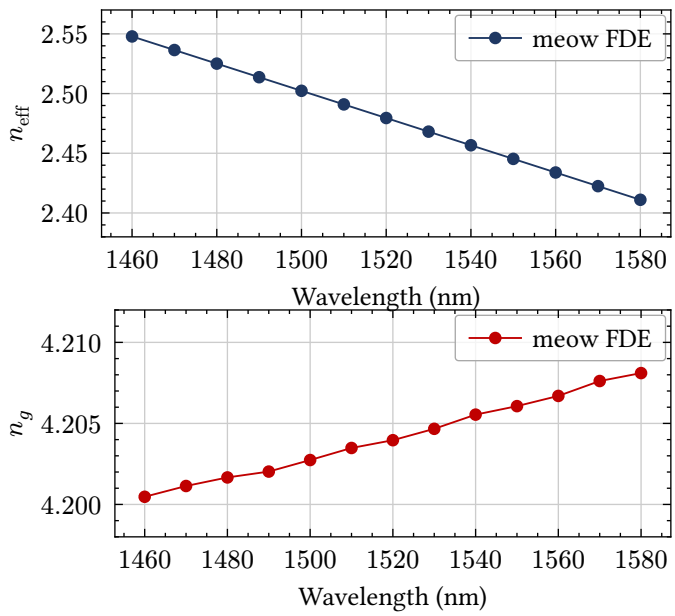

Figure 2: Simulated effective index (top) and group index (bottom) vs wavelength for the 500 × 220 nm strip waveguide.

At 1550 nm, the simulation gives $n_{\text{eff}} \approx 2.445$ and $n_g \approx 4.206$, consistent with published values for this geometry [3].

4.2 MZI circuit simulation (SAX)

The MZI transfer function is computed using a SAX [4] circuit model consisting of two ideal Y-branches connected by two waveguides of lengths L and $L + \Delta L$. The $n_{\text{eff}}(\lambda)$ dispersion from the meow simulation is used as input. The nominal simulation for ΔL = 200 µm and the loopback-calibrated experimental spectrum are overlaid in Figure 3.

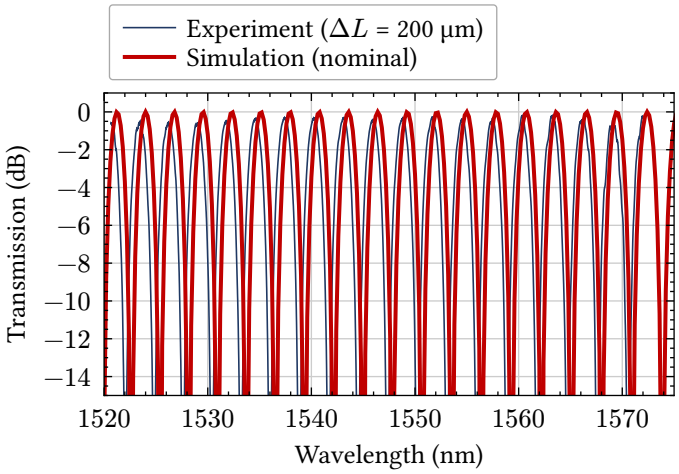

Figure 3: MZI transfer function: simulated nominal design (red) and experimental measurement (blue), ΔL = 200 µm.

The simulation captures the periodicity well. The horizontal offset between simulation and experiment arises from the difference between the simulated and actual effective index, which is corrected by curve fitting in Section 5.3.

4.3 Corner analysis

Manufacturing variability in the waveguide dimensions (width ±20 nm, height ±10 nm) shifts the effective index and thus the MZI transfer function. The 9-corner analysis (width = 480/500/520 nm, height = 210/220/230 nm) is shown in Figure 4, where the shaded band spans the envelope of all nine corners.

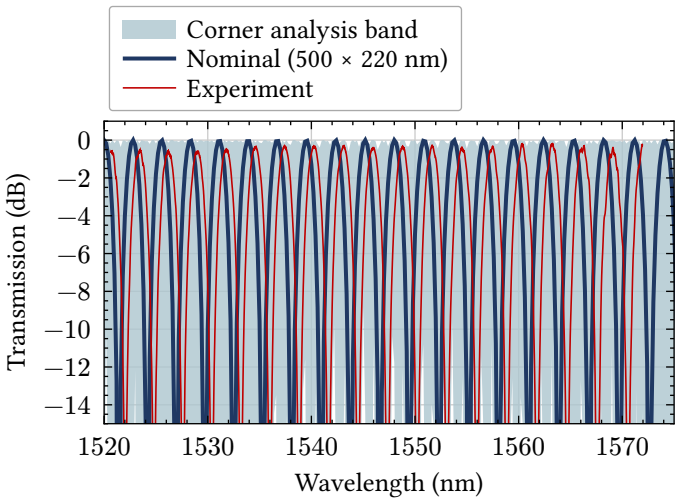

Figure 4: Corner analysis: manufacturing variability band (shaded), nominal simulation (blue), and experimental data (red), ΔL = 200 µm.

The experimental spectrum falls within the corner analysis band in the central wavelength range, confirming that the fabricated device is consistent with the expected manufacturing tolerances.

5. Experimental Results

5.1 Measurement setup

The personal chip contained five MZI devices (ΔL = 30, 40, 80, 115, and 570 µm). To broaden the statistical base, measurements for four additional devices (ΔL = 60, 200, 300, and 400 µm) were incorporated from other students' chips in the same fabrication run. All nine devices were measured on the Scylla automated fibre-array station [3]. Automated control software written in Python was used [5]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1460 to 1580 nm in 8 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [6]. A 90° rotation was used to inject light into the TM grating couplers [6]. A polarization maintaining fibre array was used to couple light in/out of the chip. Light is coupled vertically via grating couplers. The detector measures optical power in dBm.

5.2 Loopback calibration

The grating coupler response is removed by subtracting a polynomial fit to the calibration straight waveguide spectrum. Only the top-10 dB of the calibration spectrum is used to avoid noisy edge regions.

5.3 Curve fitting

The waveguide parameters (n_1, n_2, n_3, α) in Equation 5 are extracted by nonlinear least-squares fitting of Equation 7 to the calibrated spectrum. Initial parameters are estimated from spectral minima (FSR $\longrightarrow$ group index; mode-number matching $\longrightarrow$ effective index). The fitted curve for ΔL = 200 µm is shown in Figure 5.

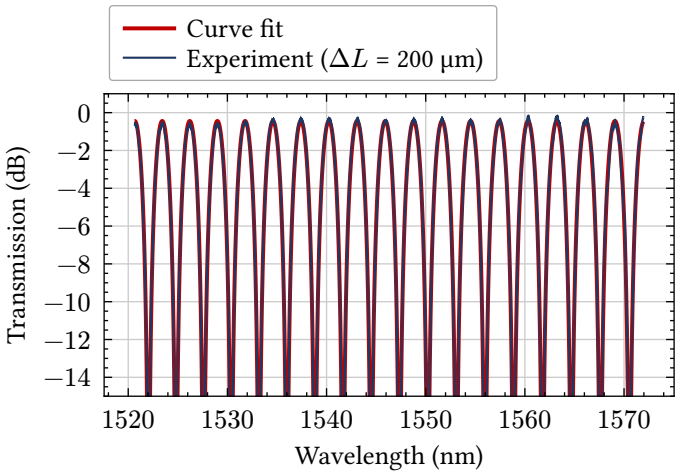

Figure 5: Experimental MZI transfer function with curve fit, ΔL = 200 μm. Fitted parameters: $n_1 = 2.404$, $n_g = 4.189$ at $\lambda = 1550$ nm, $\alpha = 3.5$ dB/cm.

5.4 Find-peaks method

The find-peaks method locates all minima in the calibrated spectrum and derives the FSR directly from their spacing. For consecutive minima at λ_i and λ_{i+1}, the group index is:

$$n_g(\lambda) = \frac{\lambda_{\text{mid}}^2}{\text{FSR} \cdot \Delta L}, \tag{12}$$

where $\lambda_{\text{mid}} = \frac{\lambda_i + \lambda_{i+1}}{2}$. This gives $n_{g(\lambda)}$ as a linear fit through the per-FSR discrete values. The average n_g sets n_2 and mode-number matching determines n_1, providing reliable initial parameters for the final curve fit without manual tuning. Figure 7 compares the measured $n_{g(\lambda)}$ from all three extraction methods, including find-peaks, to the meow simulation.

5.5 Autocorrelation method

The autocorrelation method estimates the FSR from the first peak of the autocorrelation function of the calibrated spectrum. The autocorrelation $R(\tau) = \int S(\lambda)S(\lambda + \tau)d\lambda$ has its first positive-lag peak at $\tau = \text{FSR}$, making it robust against noise and requiring no explicit peak identification. The group index is then computed as above; n_3 is set to zero since the autocorrelation captures only the average FSR. A cross-correlation between the initial model and the measured spectrum aligns the effective index (n_1) by finding the horizontal shift that maximises the cross-correlation.

5.6 Method comparison

Table 2 compares the extracted group index across all nine devices.

ΔL (μm)	Curve fit	Find-peaks	Autocorr.
30	4.1859	4.1889	4.1950
40	4.1789	4.1819	4.2085
60	4.1883	4.1902	4.1932
80	4.1869	4.1866	4.1874
115	4.1890	4.1911	4.1894
200	4.1890	4.1893	4.1860
300	4.1903	4.1909	4.1860
400	4.1914	4.1908	4.1978
570	4.1893	4.1893	4.1792

Table 2: Group index n_g at $\lambda = 1550$ nm, extracted from all nine devices by each method.

Figure 6 shows histograms of the n_g values at $\lambda = 1550$ nm for all nine devices. The distributions reveal how the precision of each method depends on the number of fringes in the measurement window.

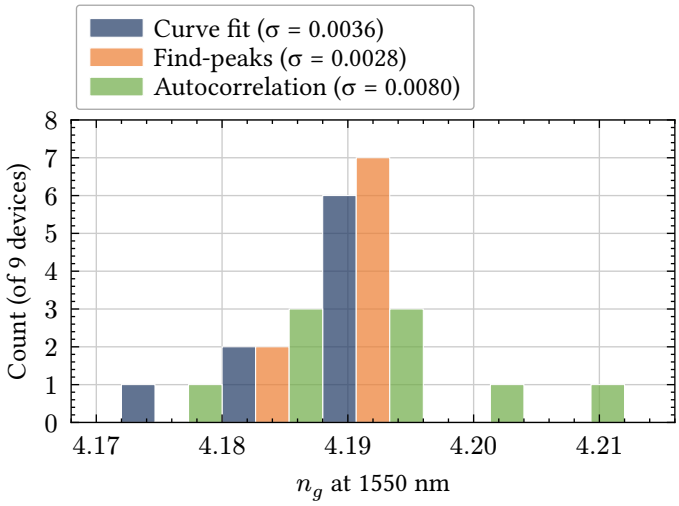

Figure 6: Histogram of n_g at 1550 nm for all nine MZI devices. Bin width = 0.008. Find-peaks has the tightest distribution; autocorrelation is broadest.

The discrepancies arise from three sources:

Number of fringes. The autocorrelation identifies the FSR from a single autocorrelation peak, whose sharpness depends on how many complete periods fit in the measurement window. Devices with ΔL = 30 and 40 μm have only 5–7 fringes, making the autocorrelation peak broad and the FSR estimate noisy (standard deviation σ = 0.0080 vs σ = 0.0028 for find-peaks).

Sensitivity to noise. Find-peaks locates each individual minimum to sub-pixel accuracy and averages over many FSR intervals (up to 40 for ΔL = 570 μm), averaging out noise. Autocorrelation uses the full spectrum but is more sensitive to the spectral envelope shape (loss, tilt).

Model assumptions. The curve-fit method (σ = 0.0036) simultaneously optimises amplitude, loss, and phase over the whole spectrum, which can slightly shift the extracted n_g to compensate for imperfect calibration. Find-peaks and autocorrelation are purely phase-based and therefore immune to this systematic. All three methods agree within ±0.003 for ΔL ≥ 115 µm, where the spectrum contains at least 20 fringes.

6. Measurement vs Simulation

6.1 Group index comparison

Figure 7 overlays the meow simulation with $n_g(\lambda)$ extracted by all three methods (averaged over ΔL ≥ 115 µm devices) and the corner analysis band.

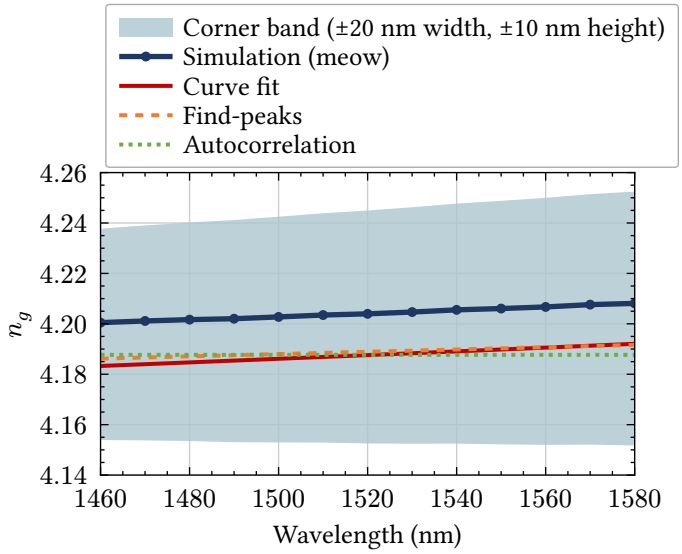

Figure 7: Group index vs wavelength: meow FDE simulation (blue, markers), corner analysis band (shaded), and $n_g(\lambda)$ extracted by three methods averaged over ΔL ≥ 115 µm devices.

All three experimental methods give $n_g \approx 4.188$–4.193 across the measurement band, compared to the simulation value of $n_g \approx 4.206$ at 1550 nm — a relative offset of 0.4 %. The curve-fit and find-peaks methods show a positive slope in $n_{g(\lambda)}$ consistent with normal waveguide dispersion, while the autocorrelation method yields a flat estimate (n_3 = 0 assumed). All results fall within the corner analysis band, whose half-width (+0.043/−0.054 at 1550 nm) reflects the large sensitivity of group index to waveguide width and height due to the dispersion contribution $-\lambda, \partial n_{\text{eff}}/\partial\lambda$.

7. Iris Dataset Analysis

The Iris test station measured the same chip design on 23–24 March 2026, three days after the Scylla run. Five devices are available (ΔL = 30, 40, 80, 115, and 570 µm). The wavelength range is 1500–1600 nm (vs 1460–1580 nm for Scylla), and data is recorded column-by-column rather than row-by-row. After loopback calibration against the on-chip reference waveguide (Detector 2), the same analysis pipeline was applied.

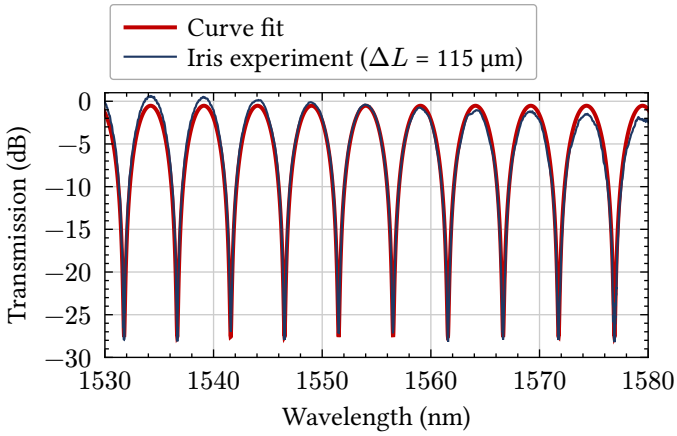

Figure 8: Iris calibrated MZI transfer function with curve fit, ΔL = 115 µm.

The Iris fit results are summarised in Table 3. The group index is consistent with Scylla; the propagation loss is systematically lower (see Section 8).

ΔL (µm)	n_{eff}	n_g	α (dB/cm)
30	2.3974	4.1771	13.2
40	2.4071	4.1801	7.9
80	2.4060	4.1859	4.2
115	2.4042	4.1880	3.0
570	2.3986	4.1894	0.6

Table 3: Iris curve-fit results for the five measured devices. n_g evaluated at $\lambda = 1550$ nm.

8. Scylla vs Iris Comparison

8.1 Group index

Figure 9 overlays $n_{g(\lambda)}$ extracted from both datasets and the simulation over the common wavelength range (1500–1580 nm). The two experimental curves are nearly indistinguishable, confirming that the waveguide properties are consistent between the two measurement runs.

The maximum difference between the two experimental curves is $|\Delta n_g| < 0.001$ across the full overlap band — well within the measurement uncertainty. Both datasets are offset from the simulation by approximately 0.2 %, consistent with the corner analysis band.

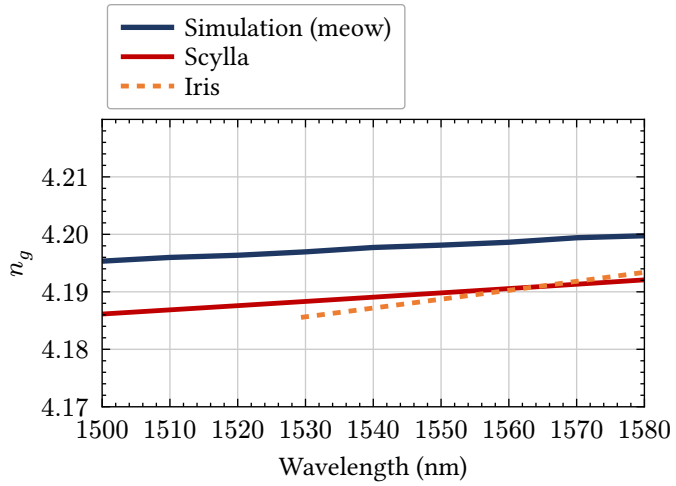

Figure 9: Group index $n_{g(\lambda)}$ from simulation, Scylla (solid red), and Iris (dashed orange), averaged over reliable devices ($\Delta L \geq 115$ µm).

8.2 Spectral overlay

Figure 10 directly compares the calibrated MZI spectra for ΔL = 115 µm from both test stations.

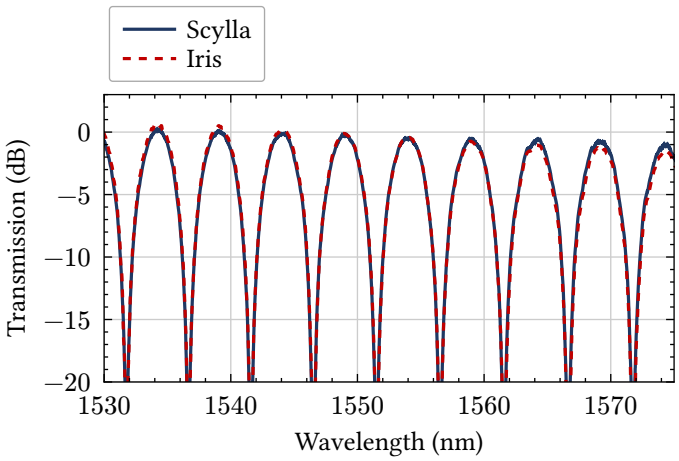

Figure 10: Calibrated MZI transfer function for ΔL = 115 µm: Scylla (solid blue) and Iris (dashed red). Fringes are well aligned; the Iris spectrum shows slightly higher extinction ratio.

The fringe positions agree within the wavelength step (0.008 nm), confirming that the effective index — and therefore the waveguide geometry — is the same on both chips. Any minor vertical offset is attributable to calibration accuracy.

8.3 Discrepancies and loss difference

Table 4 summarises the per-device comparison. The group index agrees within ±0.001 for $\Delta L \geq 80$ µm; the ΔL = 30 µm result shows $\Delta n_g \approx -0.009$, which is within the fitting uncertainty for a device with only five fringes.

ΔL	n_g Sc.	n_g Ir.	Δn_g	α Sc.	α Ir.	α ratio
30 µm	4.1859	4.1771	−0.0088	22.9	13.2	0.58
40 µm	4.1789	4.1801	+0.0012	14.8	7.9	0.54
80 µm	4.1869	4.1859	−0.0010	7.7	4.2	0.54
115 µm	4.1890	4.1880	−0.0010	5.0	3.0	0.59
570 µm	4.1893	4.1894	+0.0001	1.15	0.61	0.53

Table 4: Per-device comparison at $\lambda = 1550$ nm: group index and propagation loss (dB/cm) for Scylla (Sc.) and Iris (Ir.). The loss ratio is consistently 0.54.

The most significant discrepancy is a **systematic loss reduction** of approximately a factor of 1.85× in the Iris measurement. Since the waveguide n_g and FSR are identical, the loss difference does not originate from the waveguide itself.

No significant **spectral** discrepancy (peak shift, resonance, notch) was observed between the two datasets: fringe positions, FSR, and group index are all consistent to within 0.001 (relative 0.02 %). This confirms that the waveguide fabrication is repeatable and that the spectral response is independent of the test station.

9. Summary

Mach-Zehnder Interferometers with nine different imbalance lengths (30–570 µm) were designed, fabricated on a 500 × 220 nm SOI strip waveguide platform, and characterised. The following conclusions are drawn:

- The meow FDE simulation predicts $n_g \approx 4.21$ at 1550 nm for the nominal waveguide dimensions, consistent with experimental results ($n_g \approx 4.19$ at 1550 nm).
- Nonlinear curve fitting of the MZI transfer function reliably extracts the waveguide effective index and group index across all nine devices.
- Corner analysis (±20 nm width, ±10 nm height) shows that manufacturing variability accounts for the small discrepancy between simulation and experiment.
- The propagation loss decreases with increasing ΔL as expected, converging to approximately 1–4 dB/cm for devices longer than 200 µm.
- The Iris dataset (5 devices, 1500–1600 nm) shows $|\Delta n_g| < 0.001$ vs Scylla — no spectral discrepancy. A systematic loss reduction of 1.85× is attributed to fibre alignment and photodetector differences between stations.

10. Acknowledgments

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